Midterm Exam 1
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Question 1.

a. Determine the ranges of x for each class

Answer:
Class 1 range = (0 < x < e^2.66)
Class 2 range = (e^2.66 < x < e^6.8057)
Class 3 range = (e^6.8057 < x)

X = e^u

Since priors are all equal, and must add up to 1, all priors = .3333 (⅓)

The function simplifies to: $\frac{1}{\text{Sigma}}$ exp^(-ln(x)-muk)^2)/ 2sigma^2)

Take the log →

-ln sigma - ((lnx-muk)^2/2sigma^2)

Decision boundaries
Class 1 and Class 2→ -3.33 < x < 2.66
Class 2 and Class 3—> -.0057< x < 6.8057

Sigma = 1 and Sigma^2 = 1, therefore:

δ(class 1) $-\ln(1) - \frac{(x-1)^2}{2} = \frac{-(x-1)^2}{2}$

δ(class2) $-\ln(2) - \frac{(x-5)^2}{8}$

δ(class3) $-\ln(3) - \frac{(x-7)^2}{18}$

δ(class1) = δ(class2) —->

$\frac{-(x-1)^2}{2}$ = $-\ln(2) - \frac{(x-5)^2}{8}$ —> solve for x

Multiply by 8 —> -4x^2 +8x -4 = -5.545-x^2+10x-25 —> move all x^2 to one side
-3x^2 + 8x - 4 = -5.545 + 10x - 25—> move all x to one side and constants to the other side
-3x^2 -2x = -26.545 →write in terms of quadratic formula
3x^2 + 2x - 26.545 = 0
A =3 , b = 2, c = =26.545

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ → Quadratic formula

X = (-2 +/- √(4-4*3*-26.545)) / 6

X = (-2 +/- 17.96) / 6

X = 2.66 or -3.33
—---

$-\ln(2) - \frac{(x-5)^2}{8}$ = $-\ln(3) - \frac{(x-7)^2}{18}$ ---> solve for x

Multiply by 72 —> -72(ln 2) - 9(x-5)^2 = -72(ln 3) - 4(x-7)^2 → simplify

>> -9(x^2 - 10x + 25) = 9x^2 +90x -225

>> -4(x^2-14x +49) = -4x^2 +56x - 196

-72 (ln2) - 9x^2 +90x -225 = -72(ln3) -4x^2 +56x - 196 —> combine similar terms

5x^2 - 34x + 29 - 72(ln3) + 72(ln2) = 0 —> break down the logs

>> -72(ln3) + 72(ln2) = -72(ln3-ln2) → -72ln(3/2) = -29.1935

5x^2 - 34x -.1935 = 0
A = 5, b = -34, c= -.1935

Solve in quadratic equation —>
X = -0.0057 and 6.8057

Question 2:

a) **Write down log odds model for logistic regression model.**

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ind	X1	X2	y
1	1	-1	1
2	1	1	1
3	-1	1	0
4	-1	-1	0

Odds = beta0 · |X1|^beta1 · |X2|^beta2
Answer: Log odds = log(β0) + β1·log|X1| + β2·log|X2|

b) **Likelihood function of the dataset**

L = BIG PI{where y=1} pi_i * BIG PI{where y=0} (1 - pi_i)
L = pi_1 * pi_2 * (1 - pi_3) * (1 - pi_4)
p1= [beta0 * |1|^beta1 * |-1|^beta2] / (1 + [beta0 * |1|^beta1 * |-1|^beta2])

Since abs(-1,1) all = 1 and ^ any power = 1, then:

All rows will simplify to: p = beta0/(1+beta0)

Answer: L = pi^2 * (1-pi)^2

c) **Parameters that find the likelihood function**

p(y=1) = .5
p(y=0) = .5
Equal probability
P = beta0/(1+beta0)
p(1+beta0) = beta0
Which simplifies to:
.5 + .5beta0 = beta0
.5 = beta0 - .5 beta0 (.5 is half of beta0, therefore multiply by 2 to get beta0)
Beta0 = 1

Answer: Beta0 is the only parameter that remains in Likelihood function, and is equal to 1.

Question 3.
Choice E (Lasso) is a valid regularizer that drives beta coefficients that are 0 in value, driving towards a sparse model with an optimal number of effective coefficients. **Choice B** (Log penalty) is a valid regularizer that can yield sparse model, but unlike Lasso, it does not overly penalize large coefficients. Choice A and Choice F are technically valid as they both the coefficients exponentially and penalize larger coefficients, but neither drive small coefficients to 0 and do not yield sparse models. Since we do not regularize beta0, Choice C is not a valid regularizer.

Question 4.
Classify using prior
N = 9
Counts(+) = 4
Counts(-) = 3
Counts(*) = 2

CV = 1/9 * SUM (MSE1, MSE2,...MSE9)

Priors when all sets used (n=9)
P(+) = 4/9 = 0.444
P(-) = 3/9 = 0.333
P(*) = 2/9 =0.222

Winning class when {index} is left out:
Using LOOCV → using 'V' as validation set and 'T' as training set, Y_pred is the winning class based on T. Since y class is qualitative, I'm calculating error based on binary class using right = 1 and wrong =0.

V	T	Y_pred	Y actual	Right/wrong
1	2,3,4,5,6,7,8,9	+	+	1
2	1,3,4,,5,6,7,8,9	+	-	0
3	1,2,4,5,6,7,8,9	+	-	0
4	1,2,3,5,6,7,8,9	+	*	0
5	1,2,3,4,6,7,8,9	+	-	0

6	1,2,3,4,5,7,8,9	+	+	1
7	1,2,3,4,5,6,8,9	+	+	1
8	1,2,3,4,5,6,7,9	+	*	0
9	1,2,3,4,5,6,7,8	+	+	1

Answer: Total misclassified using LOOCV = 5/9 =0.556

Question 5.
C1 = alligator, Mu(c1) = 12
C2 = crocodile, Mu(C2) = 15
SD = 2 feet

a. Conditional pdf of X given being a crocodile

$= f(x \mid C2) = \frac{1}{2\sqrt{2}p(class2)} \wedge (-(x-15)^2 / 8)$

Since $2*SD^2 = 2 * 2 * 2 = 8$

Since SD(C1) = SD(C2) = 2 = equal, this simplifies to: $x * (15/4) - (15^2/8) + \ln(pcroc)$

$f(x \mid C2) = x * 3.75 - (225/8) + \ln(prcroc) = x * 3.75 - 28.125 + \ln(prcroc)$
Conditional PDF = $3.75x - 28.125 + \ln(prcoc)$

b. Conditional pdf ov X given being an alligator

$= f(x \mid C2) = \frac{1}{2\sqrt{2}p(class1)} \wedge (-(x-12)^2 / 8)$

$x * (12/4) - (12^2/8) + \ln(pallig)$

$f(x \mid C1) = x * 3 - (144/8) + \ln(pallig)$

Conditional PDF = $3x - 18 + \ln(pallig)$

c. Prior probability same

Given p(C1) = p(C2) = .5

P(Croc | X = *) = P (Allig | X = x*)

Decision boundary = $\frac{(15+12)}{2} + \frac{4}{3} \ln(.5/.5)$

Since .5/.5 = 1 and ln(1) = 0, then = 27/2 is decision boundary

Answer: 13.5 feet is the decision boundary, so any animal over 13.5 feet will be classified as a crocodile, and any animal less than 13.5 feet will be classified as an alligator

d. Population of crocs twice as much as population of alligators

P(Croc) = 2x Prob(Allig)

P(Croc)/2 = P(Allig) and P(Croc) + P(Allig) = 1 (We know priors add up to 1)
1 = 2xProb(Allig) + 1xProb(Allig) → since we know Prob(croc) = 2Prob(Allig)
Then Prob(Allig) = ⅓ =0.333
And Prob(Croc) = 2/3 = .66667

.666/.333 = 2

So decision boundary = 13.5 - 4/3 * ln(2)
Since ln(2) = .6931, then 13.5 - 1.333 * .6931
Answer = 12.58 ft is the decision boundary, which means if it is greater than 12.58, it is classified as a crocodile

e. Population of alligators twice as much as crocs

P(Allig) = 2xProb(Croc)

= 13.5 - 1.333 * ln(½) =
Since ln(.5) = -.6931, then 13.5 +.9239 = 14.424
Answer = 14.42 feet is the decision boundary, which means animals less than 14.42 feet are classified as alligators, while the remaining are classified as crocodiles.

f. sd different
sd(croc) = 4
sd(allig) = 2

Croc
sd(croc) = 2xsd(allig)

Crocodile | Alligator

$\frac{(x-15)^2}{2*16} - \log(4) + \log(pi_croc) = \frac{(x-12)^2}{2*4} - \log(2) + \log(pi_allig)$

Since it's given that priors are same, log(pi_croc) = log(pi_allig) then we can remove from each side of equation:

$\frac{(x-15)^2}{32} - \log(4) = \frac{(x-12)^2}{8} - \log(2)$

$\frac{(x-15)^2}{32} = \frac{(x-12)^2}{8} + \log(2)$ —> multiply both sides by 32

(x-15)^2 = 4(x-12)^2 - 32log(2) —-> 32 * log(2) = 22.18

(x-15)^2 simpflies to x^2 -30x +225

And 4(x-12)^2 simplifies to 4 * [x^2 -24x^2 + 144] = 4x^2 -96x^2 +576

So 31x^2 +225 = -92x^2 +576

Simplify:
123x^2 + 225 = 576 → subtract 225 from both sides
123x^2 = 351 —> divide by 123
x^2 = 2.85366

Question 6.
N = 28

a. Estimated coefficients of betahat0 and betahat1

Answer: The irreducible error is 6.15 (beta hat 0), or the random natural noise in the world that cannot be explained by any mode. For every additional meal a student eats per week, we can expect their weight to increase by .553 pounds (beta hat 1) assuming other variables constant.

b. Alpha = .01 // H0 = beta_3 = 0

t = (beta_3 -0)/ (SE(beta3) → since beta_3 is .813 and beta_3 SE is .202, then:

Simplifies to = .813/.202 = 4.02475

Null H → beta_3 is 0
Alt H → beta_3 is not 0
Decision Rule → P-value < or = alpha —> P-value < or = .01

Degrees of freedom = n- k (predictors) - 1 → 28 - 3-1 = 24

Critical value = 2.797 (Based on 24 degrees freedom, alpha/2 = .005)
Answer: Since the t-statistic 4.02 is greater than the critical value of 2.797, we reject the null hypothesis that the beta_3 coefficient is = 0.

c. Find the 94% CI for beta_1
Alpha = .06 and alpha/2 = .03

T statistic at alpha/2 = .03
Dof = 24

T-critical value at (alpha/2=.03) with 24 degrees of freedom is 1.9740
Therefore Margin of error = 1.9740 * 0.165 = 0.32571
Upper bound = 0.553 + 0.32571
Lower bound = 0.553 - 0.32571

Answer: We can be 94% confident that the true value of beta_1 falls between 0.2273 and .9787

D. perform F-test for overall significance
Numerator degrees of freedom = k = 3
Denominator degrees of freedom = n - 3-1 = 24
Alpha = .03

H0 (Null Hypothesis): Beta_1 = Beta_2 = Beta_3 = 0

Alternative hypothesis: At least one Beta (1,2,3) does not = 0

$$F = \frac{83.4/3}{43.9/24} = 27.8 / 1.829 = 15.1996$$

F-distribution of alpha = .03 with df (3,24) = 3.5304

Answer: Since our F-statistic 15.1996 is greater than the F-critical value of 3.5304 at alpha =.03, we reject the null hypothesis that all coefficients of the model are 0.There is significant evidence that at least one coefficient of the model is statistically significant overall.

Question 7.
Gaussian distribution, SD = 1

a. Best guess of Weight given Height

Mean squared error → (w-0.5 x h^1.001)^2 → w is the variable, and 0.5xh^1.001 is the mean of weight, or in this case the best guess for someone's weight in terms of height.

2sigma^2 = 200 —> so sigma^2 = 100 → sigma = 10
SD → 10 and variance = 100

Answer: w = f(h) = 0.5h^1.001

b. Estimate weights of people who's heights are 155, 165, 190

Plugging into the equation we get:

Answers
For height=155—> 0.5 x 155^1.001 = 77.89 kg
For height = 165 → 0.5 x 165 ^1.001 = 82.92 kg
For height = 190 → 0.5 x 16^ 1.001 = 95.50 kg

c. Answer: The best guess for weight as function of height would not change if the variance were also given as a function of height, because variance measures the spread, and would not change the mean of weight (0.5h^1.001). Therefore, the results for 7b would remain the same.

d. Estimate weight of people using KNN

When height=150, k = 3 → 3 closest points are Person 5(150, 65), Person 2(168, 78), and Person 1(171, 80)
Therefore Y^KNN = (65+78+80)/3 = 74.333kg (when h = 150)

When height = 155, k = 3 → KNN is Person 5, Person 2, Person 1
= 74.33 kg (when h = 155)

When height = 165 → KNN is Person 2, Person 1, Person 6 → (78 + 80+ 83)/3 = 80.333 kg (when h=165)

When height = 190cm → KNN is Person 3, Person 4, Person 6→ (83+ 80+ 100)/3 = 87.667 kg (when h=190)

e. Use weighted averages

1/d = 1/distance of

For Height = 150→

KNN is P5 (150,65) P2(168,78) and P1(171,80) so

W1 = 1/d1 = 1/0 = infinity
W2 = 1/d2 = 1/18 = 0.056
W3 = 1/d3 = 1/21 =0.048

Anwer: Since test instance height is equal to point 5 (height 150, weight 65), the estimated guess is 65 kg.

For Height = 155 →
KNN is P5, P2, P1

W1,2,3 = 1/5, 1/13, 1/16
Y1,2,3 = 65, 78, 80
So W1Y1 + W2Y2 + W3Y3 = 13 + 6 + 5 = 24
W1+W2+ W3 = ⅕ + 1/13 + 1/16 = 0.339
24 / 0.339 = 70.8 kg (for height =155)

For Height = 165 (KNN = P2, P1, P6)
W1,2,3 = 1/3, ⅙, 1/13
Y1,2,3, =78, 80, 83
So W1Y1 + W2Y2 + W3Y3 = 26 + 13.33 + 6.385 = 45.715
W1+ W2 + W3 = .577
45.715/.577 = 79.23 kg (for height = 165)

For Height = 190 (KNN = P3, P4, P6)
W1,2,3 = 1/1, ⅛, 1/12
Sum W = 1.20833
Y1,2,3, = 100, 80, 83
So W1Y1 + W2Y2 + W3Y3 = 100 + 10 + 6.9166 = 116.91667
116.91667/ 1.20833 = 96.759 kg (for height = 190cm)